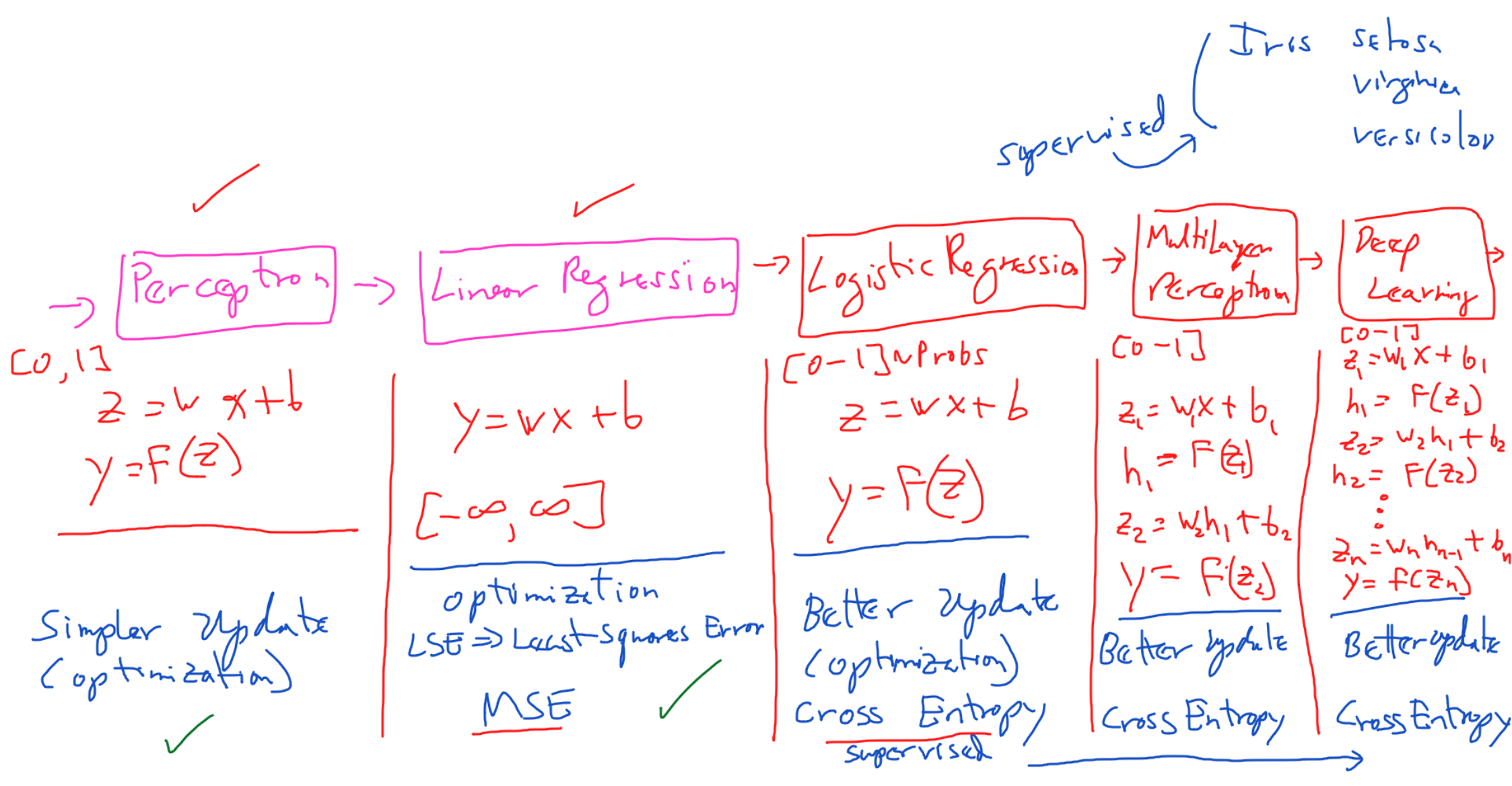
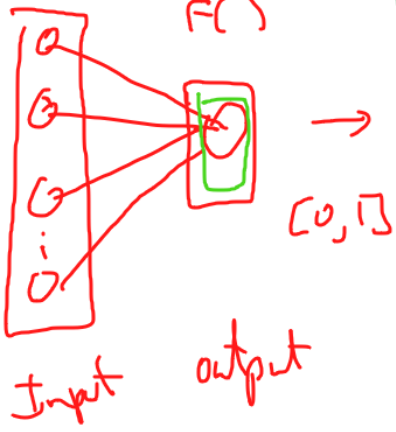
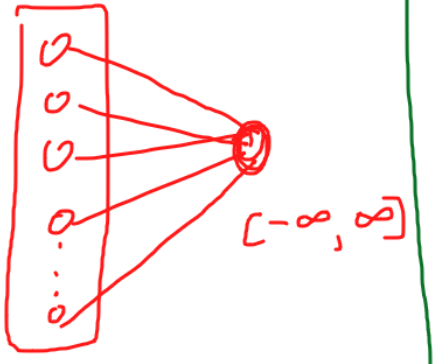

Iris
setosa
virginica
versicolor
supervised
Perceptron
Linear Regression
Logistic Regression
MultiLayer Perceptron
Deep Learning
[0,1]
z = w x + b
y = F(z)
Simpler Update (optimization)
y = wx + b
[-∞, ∞]
optimization
LSE ⇒ Least-Squares Error
MSE
[0-1] nProbs
z = wx + b
y = F(z)
Better Update (optimization)
Cross Entropy
supervised
[0-1]
z1 = w1 x + b1
h1 = F(z1)
z2 = w2 h1 + b2
y = F(z2)
Better Update
Cross Entropy
[0-1]
z1 = w1 x + b1
h1 = F(z1)
z2 = w2 h1 + b2
h2 = F(z2)
zn = wn hn-1 + bn
y = f(zn)
Better Update
Cross Entropy

Architecture

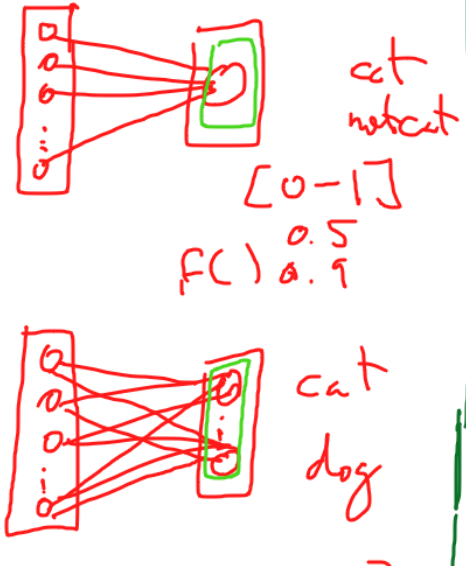
Perceptron



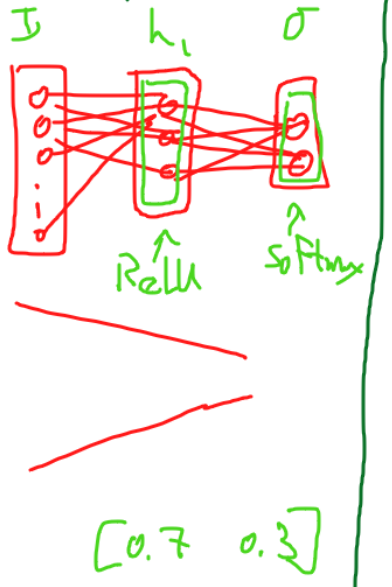
Linear Regression



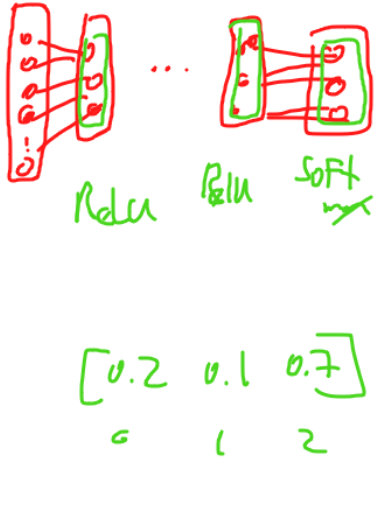
Logistic Regression



Multilayer Perception



Deep Learning



Sigmoid [0-1]

Softmax [0-1]

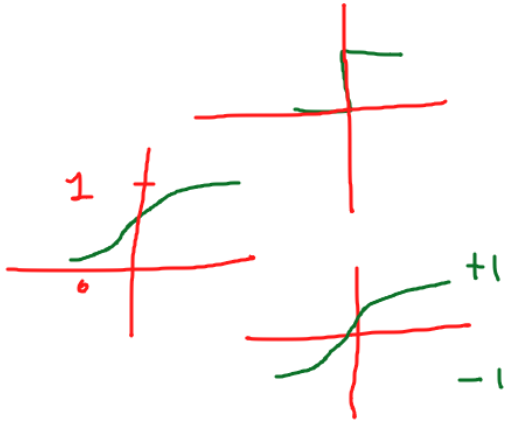
[0.8 0.2]

Activation Functions F( )

step(z)

Sigmoid(z)

tanh(z)



step(z) [0, 1]

Sigmoid(z) [0 - 1]

tanh(z) [-1 <-> +1]

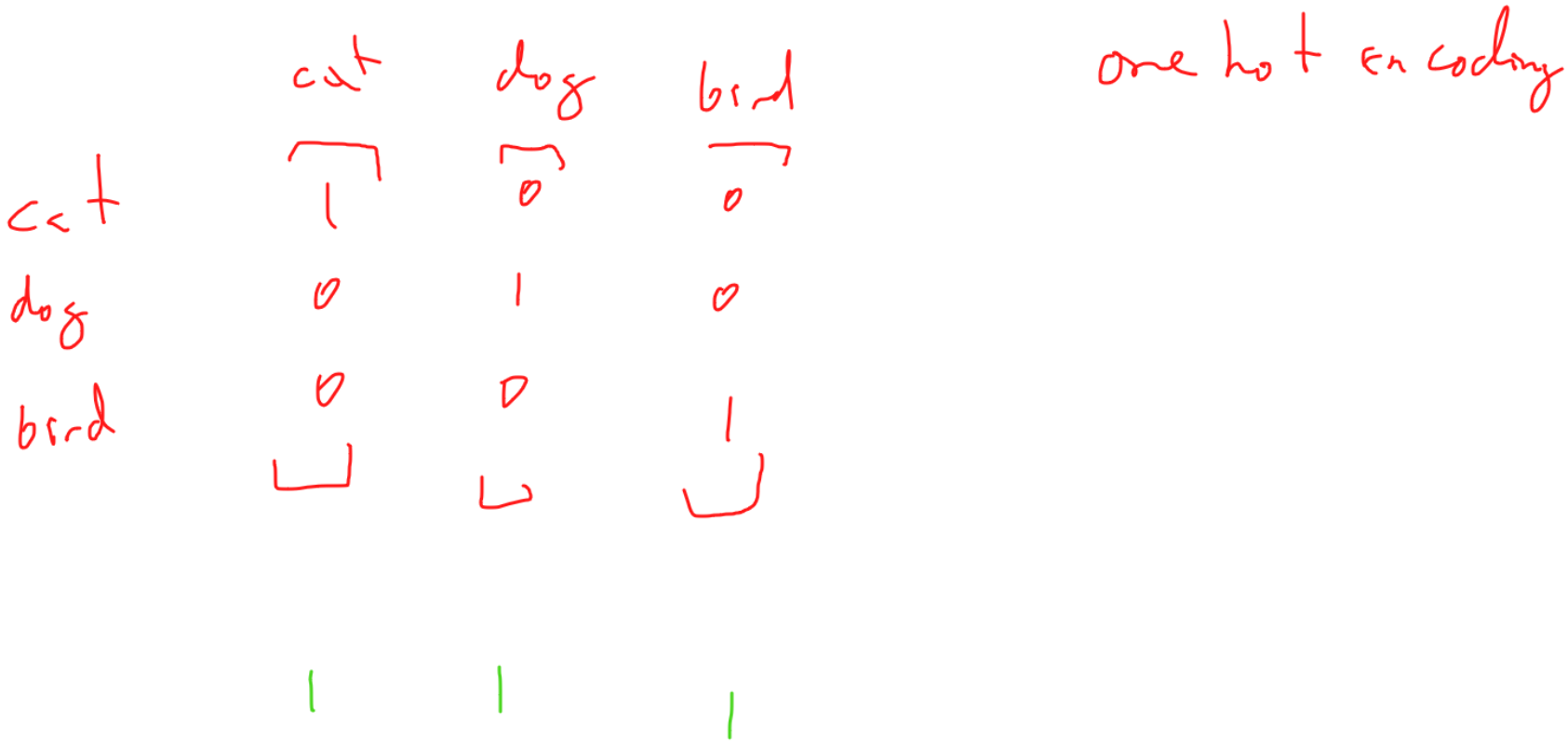
SoFtmax(z) Like Sigmoid but values add zp to 1

✓ ** ReLU (z)

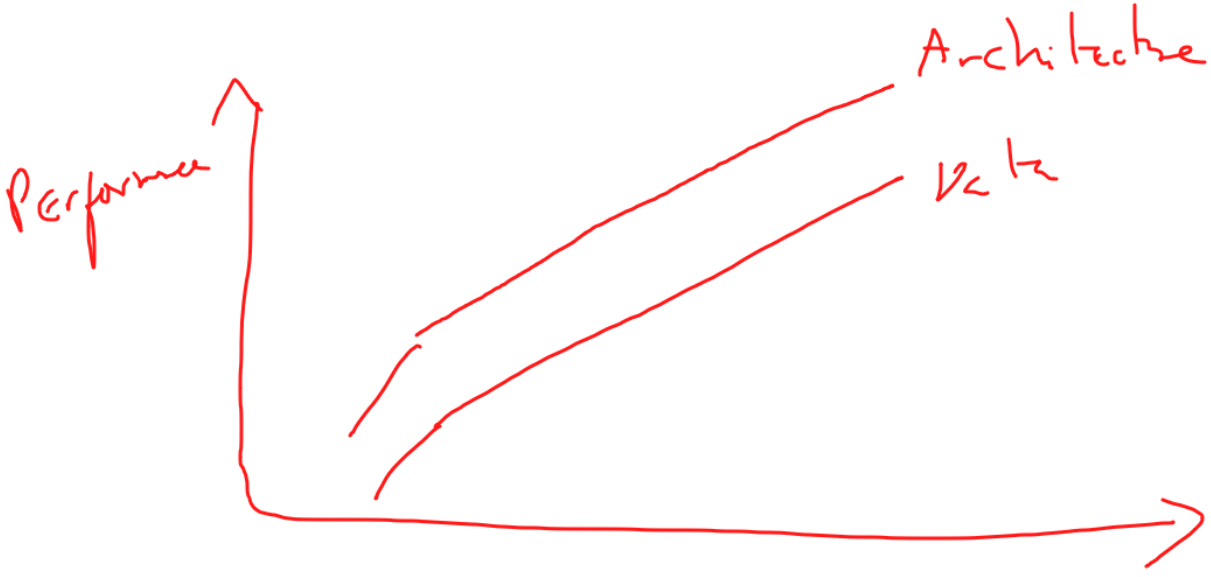
✓ GeLU (z)



[0, +∞]



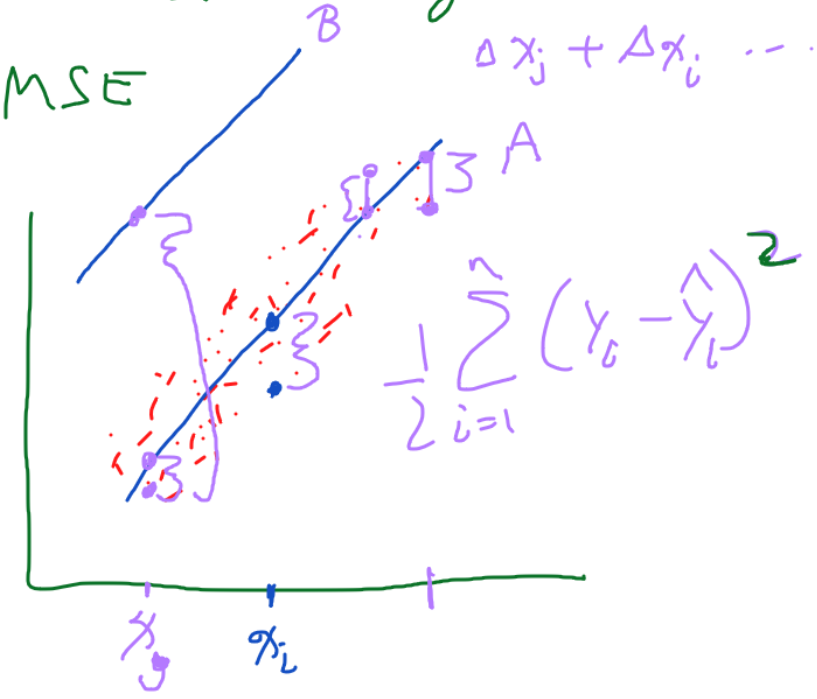
one hot encoding


Performance
Architecture
Value

Optimization (Loss Function)

Linear Regression

- MSE

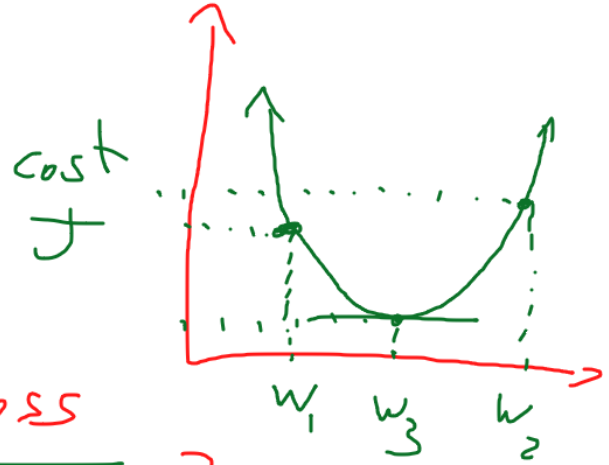


Classification (Logis Reg, MLP, DL)

- Cross entropy



Linear Regression



Inference

$$y = wx + b$$



Loss

$$J = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$J = \sum_{i=1}^{n} ((wx+b) - \hat{y}_i)^2$$

Cross Entropy (DL)

Inference

$z = \underline{w} x + b$ } more layers

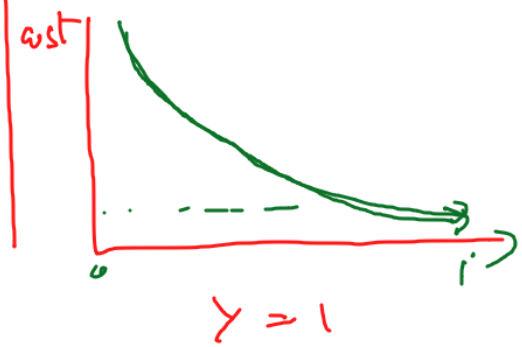
$y = F(z)$

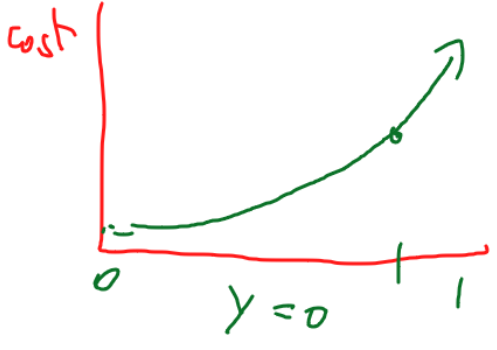
$[0-1]$

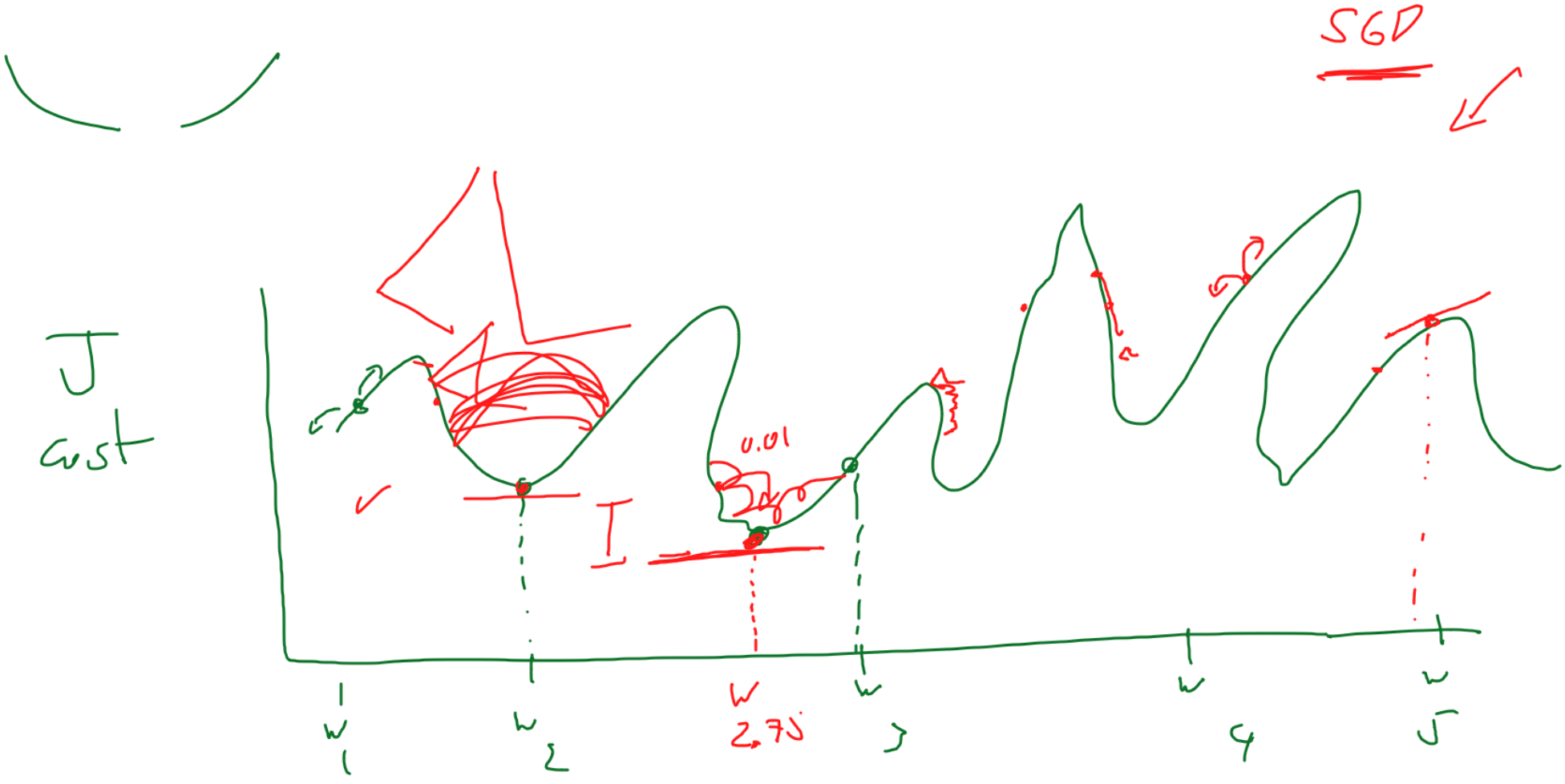
Loss (binary)

real class

$$J = \begin{cases} -\log(g(\vec{w}x+b)) \rightarrow \text{if } y=1 \\ -\log(1-g(\vec{w}x+b)) \rightarrow \text{if } y=0 \end{cases}$$






SGD
J
cost
0.01
I
W
1
W
2
W
2.75
W
3
W
4
W
5